



**TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC 2/ IP YEAR 6 2017**

CANDIDATE
NAME

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CENTRE
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INDEX
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CLASS

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H2 BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

**Thursday 14 September 2017
2 hours**

Additional materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Section A

Write your name, Centre Number, index number and class in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graph.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the separate Writing Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Q1	/11
Q2	/11
Q3	/28
Essay	/25
Total	/75

This document consists of **17** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows an example of how PCR and gel electrophoresis can be used to identify the allele that codes for the blood type antigen that is present in an individual.

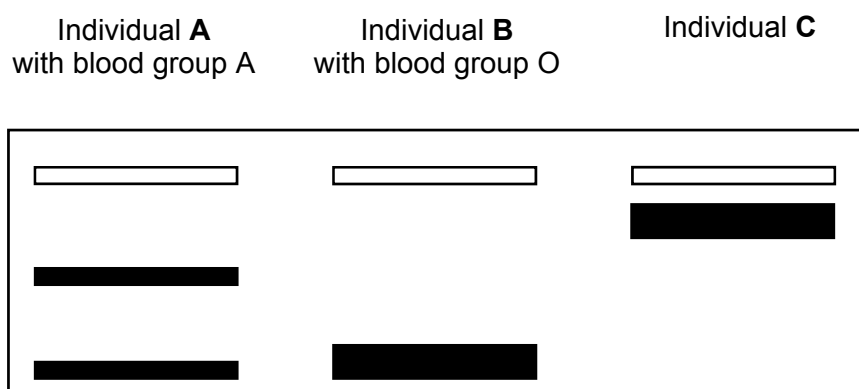


Fig. 1.1

- (a) State if individual **C** can donate blood to individual **A**. Explain your answer.

[2]

- (b) State **one** limitation of using PCR in determining the genotype of the individual.

[1]

- (c) Explain why Southern blotting can be used to confirm the presence of the allele coding for blood type A antigen.

[2]

Malaria is a vector-borne disease caused by *Plasmodium* spp.

Plasmodium gametocytes can differentiate into female and male gametes. To correctly identify male and female *Plasmodium* gametes, real-time polymerase chain reaction (PCR) was used. 60ng of DNA samples from different types of *Plasmodium* gametes were first prepared in separate tubes. Primers for a specific gene sequence were added to each tube of DNA sample. Free nucleotides tagged with fluorescent dye were also added.

Fig. 1.2 shows the level of fluorescence measured as PCR progressed.

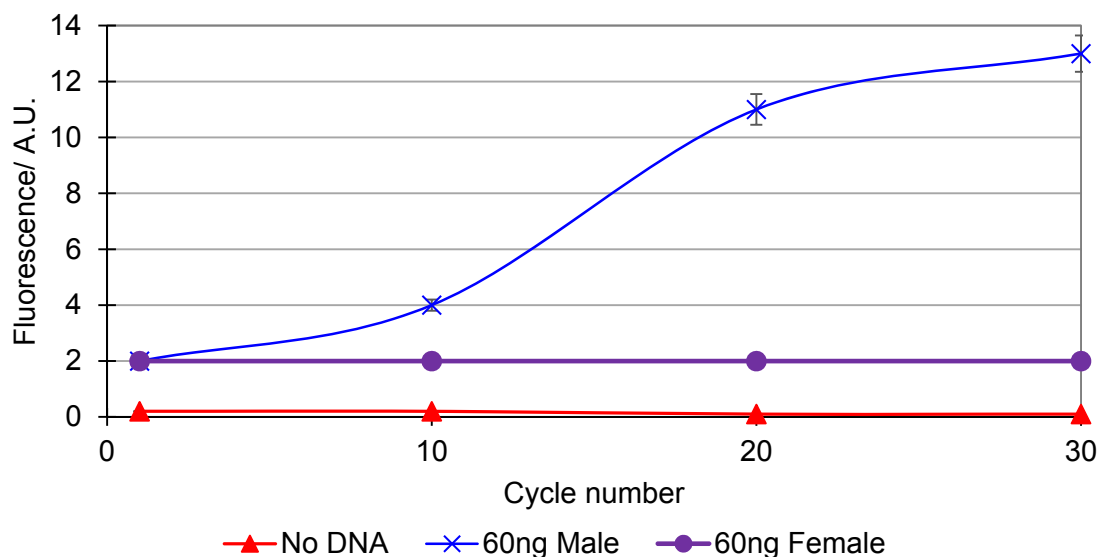


Fig. 1.2

- (d) (i) State the type of chromosome on which the gene is found on. Explain your answer.

[2]

- (ii) Predict and explain what will happen to the fluorescence level after **30** cycles for the DNA sample from the **male** gametes.

[1]

- (iii) Real-time PCR can also be used to determine the initial quantity of the sample.

On Fig. 1.2, sketch how the graph will look like when the mass of starting DNA sample is reduced to **30ng**.

[1]

- (e) Distinguish the products formed from real-time PCR with that of DNA replication.

[2]

[Total: 11]

2 Hepatitis C virus (HCV) is an enveloped, positive single-stranded RNA virus that infects hepatocytes (liver cells). The virus is attached to a low-density lipoprotein (LDL) upon its release from hepatocytes.

Fig. 2.1 shows how HCV enters and leaves a hepatocyte.

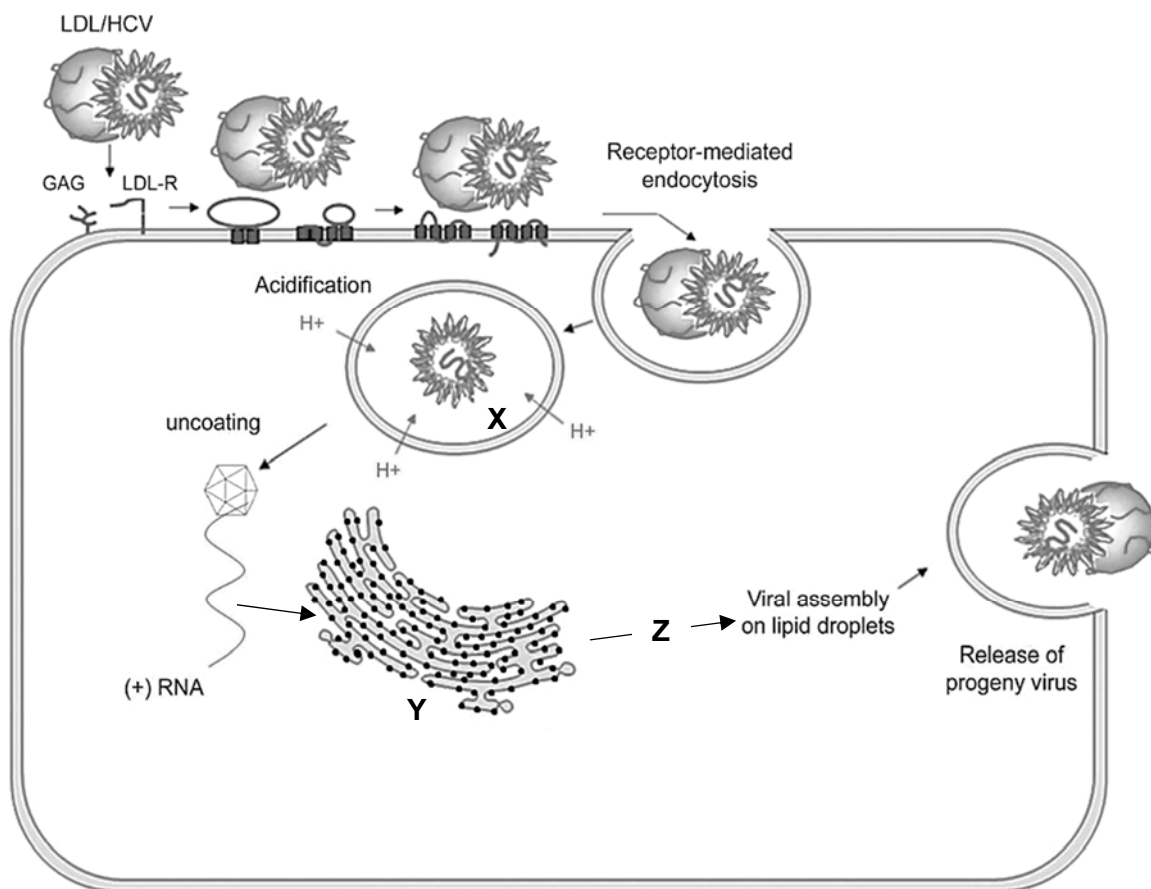


Fig. 2.1

(a) With reference to Fig. 2.1, state the role of LDL with respect to the viral replication cycle.

[1]

(b) Describe the functions of organelle X with respect to the replication cycle of HCV.

[2]

- (c) One of the organelles involved in HCV replication cycle is **Z**.
Suggest the identity of **Z** and its role in the virus replication pathway.

[2]

- (d) Explain the role of the positive single-stranded RNA as a genetic material in HCV.

[2]

The HCV core protein has been found to be oncogenic. An increase in the viral core proteins within hepatocytes can lead to the activation of cell proliferation signaling pathway.



Fig. 2.2

- (e) Define the term *oncogenic* in this context.

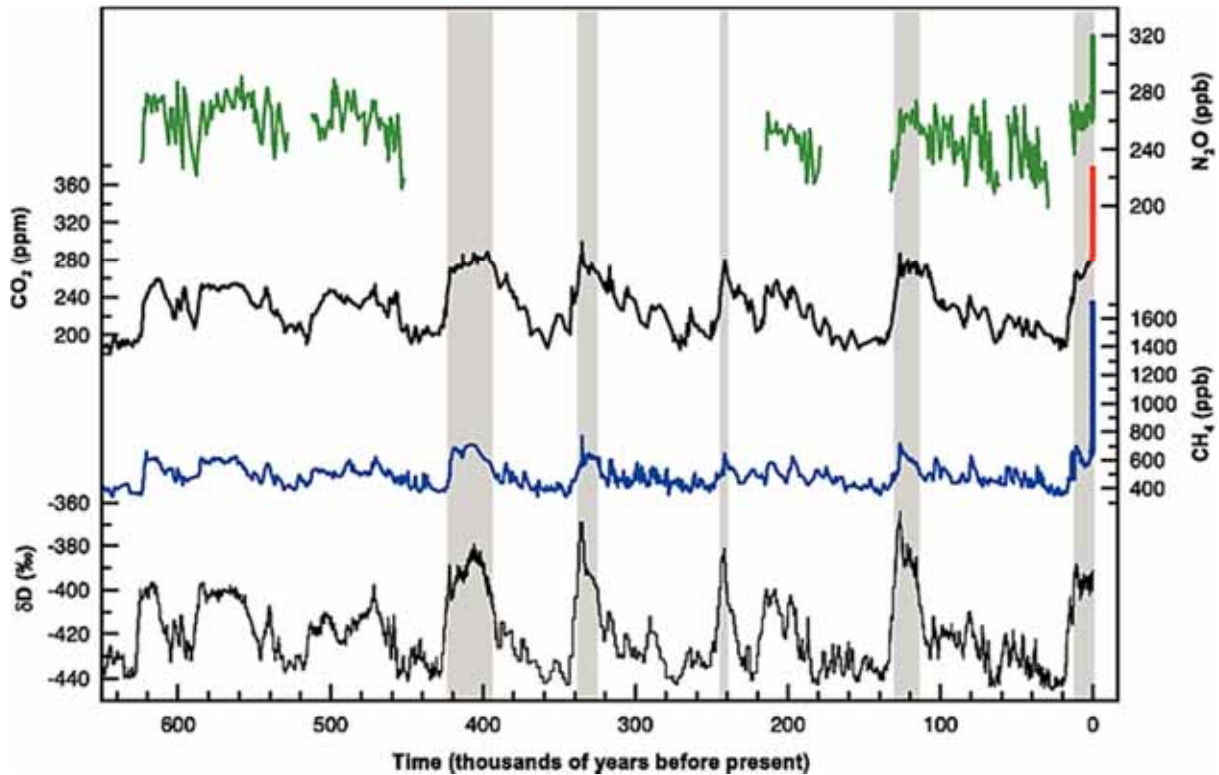
[2]

- (f) Out of 3 million individuals infected with HCV, less than 5% developed liver cancer.
Account for the low percentage of HCV infected individuals developing liver cancer.

[2]

[Total: 11]

- 3 Fig. 3.1 shows the atmospheric concentrations of carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) in the air that was trapped within ice cores. The changes in temperature is reflected as variation of deuterium (δD). The shading indicates the last interglacial warm periods.



Source: IPCC Fourth Assessment Report: Climate Change 2007

Fig. 3.1

- (a) (i) Without quoting any numerical data, state the general relationship between the greenhouse gases and temperature in Antarctica.

[1]

- (ii) Suggest the possible consequences of increased temperature on the physical environment in Antarctica.

[2]

Climate change critics argue that the climatic variation as seen in Fig. 3.1 is natural.

Fig. 3.2 shows how carbon dioxide concentration and temperature has changed since year 1000, which may be used to convince critics that the extent of climate change over the last millennium is beyond what was previously seen.

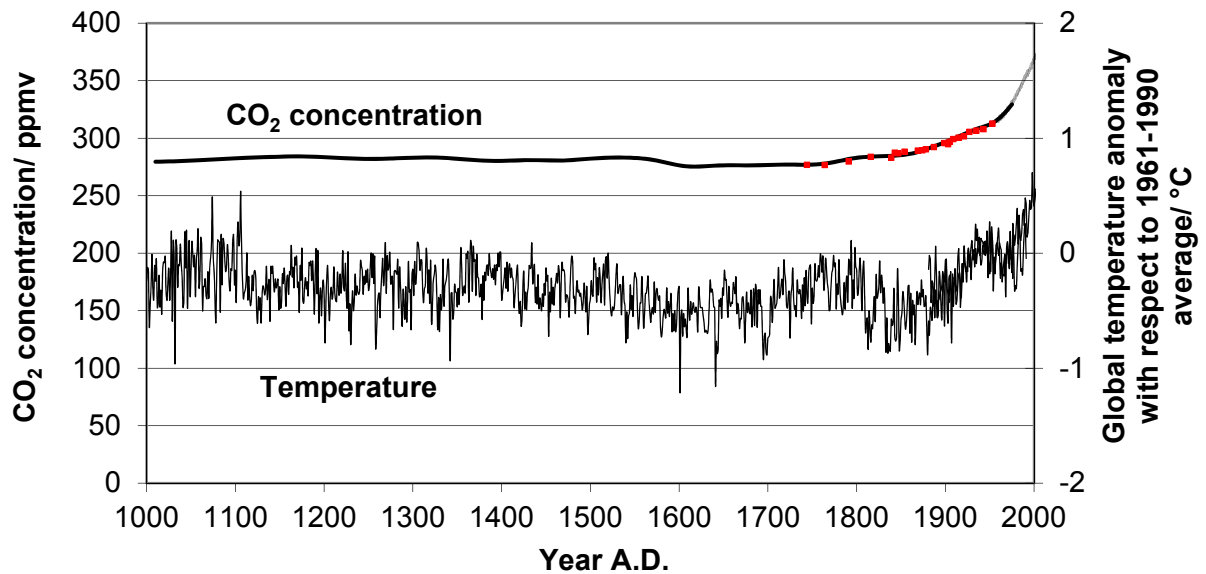


Fig. 3.2

(iii) Account for the trend of carbon dioxide concentration.

[3]

Climate change affects plants. Plants can be categorized based on the way they photosynthesize. Most plants are C₃ plants because their first photosynthetic product is a three carbon compound. Examples of C₃ plants include barley, oats and wheat commonly grown in temperate regions. On the other hand, common grass crops of tropical regions, such as maize, sorghum and sugarcane, are C₄ plants.

The rate of photosynthesis for both temperate and tropical plants were measured and shown in Fig. 3.3.

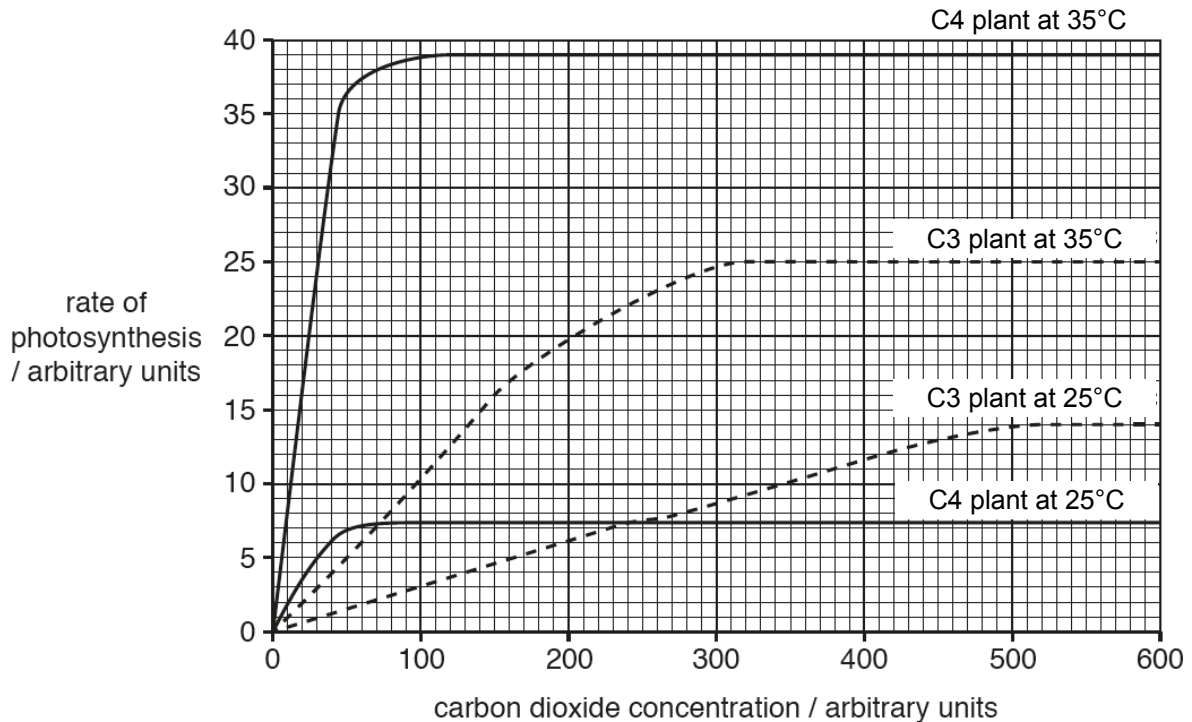


Fig. 3.3

- (b) Using evidence from Fig. 3.3, suggest which type of plant is better adapted to the impacts of climate change.

[2]

Fig. 3.4 shows the data collected on the distribution of C4 plants over time, whereby the local temperature was shown to have increased since 1940s. Three populations of the same species of C4 plants were tracked over time.

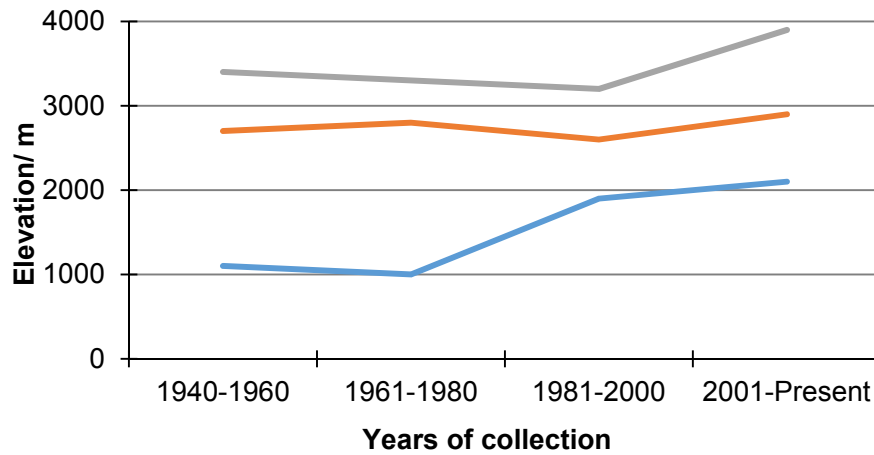


Fig. 3.4

(c) Account for the effect of increased temperature on the distribution (altitude) of C4 plants.

[2]

Fig. 3.5 shows the effects of climate change on the potential change in yield and geographical distribution of wheat and maize plants at sub-continental level from 2005 to 2050.

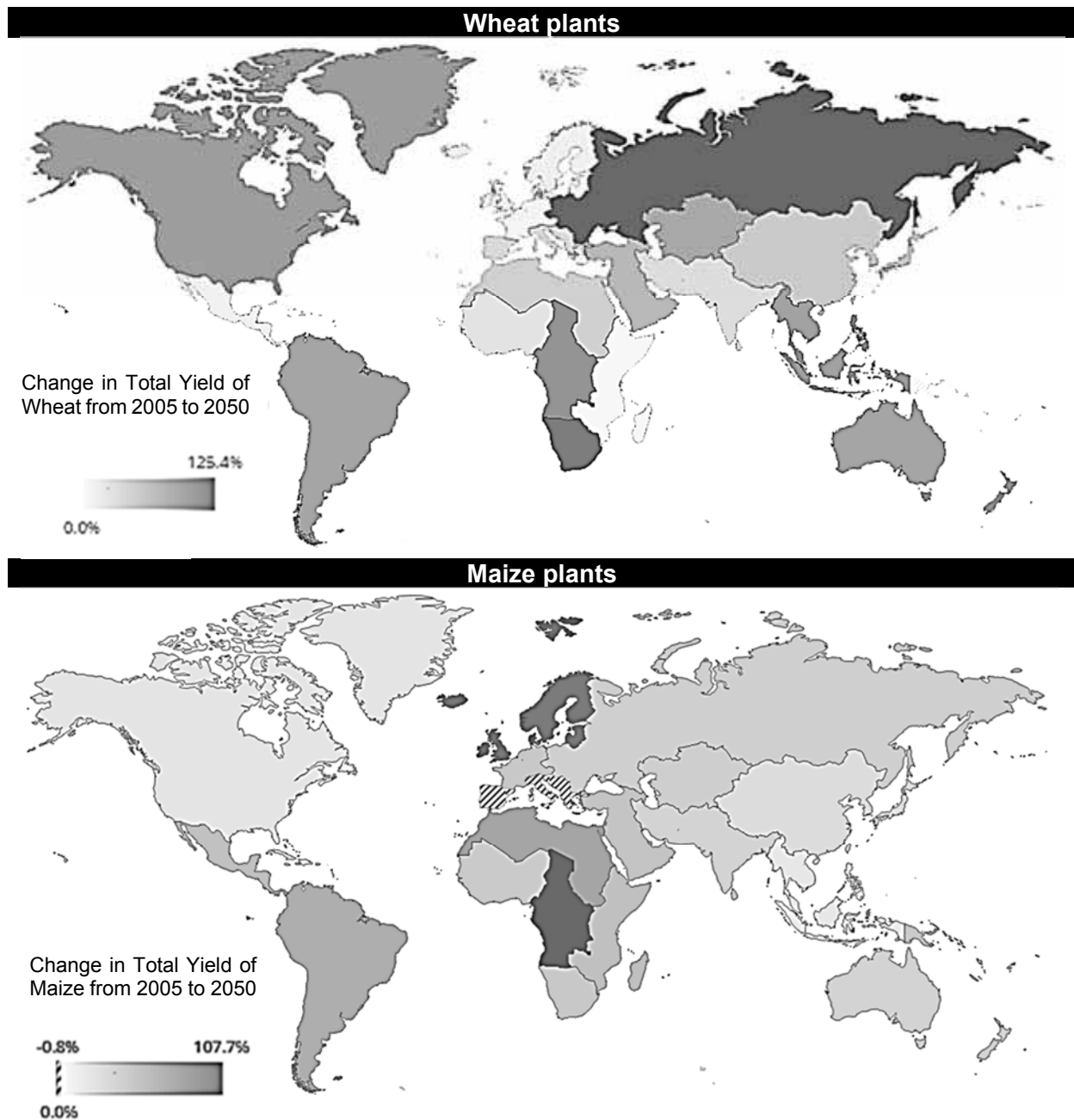


Fig. 3.5

- (d) State the possible consequence of increased temperature on global wheat and maize supply in 2050.

[1]

Climate change also has impact on the physiology of insects, thus affecting the transmission of vector-borne diseases. One such vector is the mosquito.

- (e) (i)** Outline the general life cycle of a mosquito.

[2]

- (ii)** Blood meals are a good source of protein for mosquitoes for the production of eggs.

Explain why blood is a good source of protein.

[1]

Fig. 3.6 shows the effect of temperature on the egg-to-adult survival rates and the behaviour of adult mosquitoes.

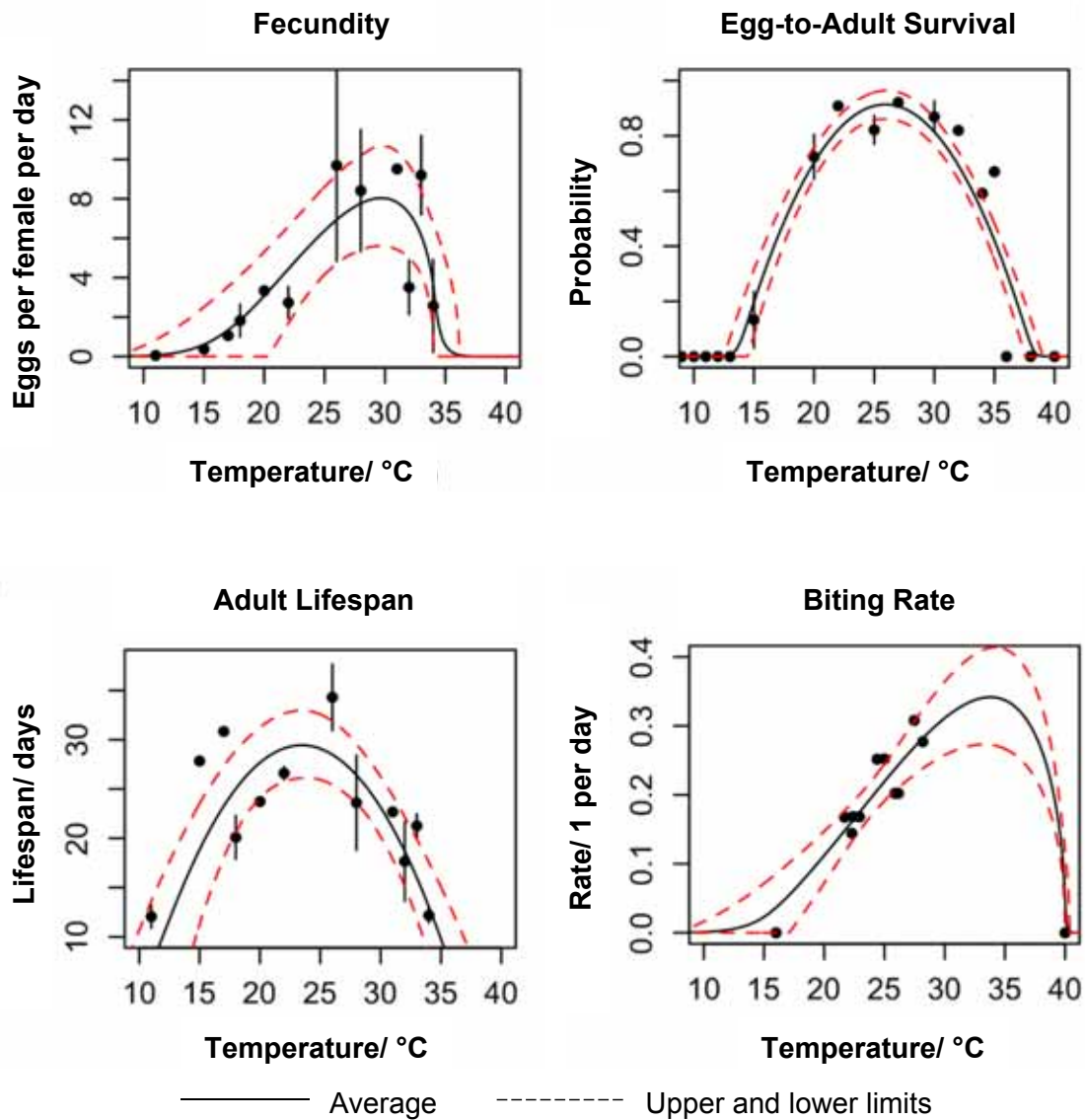


Fig. 3.6

- (f) Using evidence from Fig. 3.6, discuss the rate of transmission of diseases caused by mosquitoes when temperature increases from 30 to 34°C.

[3]

Table 3.1 describes the two types of mosquitoes that can transmit dengue.

Table 3.1

<i>Aedes aegypti</i> (Yellow fever mosquito)	<i>Aedes albopictus</i> (Asian tiger mosquito)
Found both indoors and outdoors	Found mostly outdoors (e.g. gardens)
High preference for taking blood meals from humans and to lesser extent from domestic mammals, which makes it a very capable vector of dengue viruses	Bites humans but also a variety of available domestic and wild vertebrates that do not carry the dengue viruses, which lowers its capacity to transmit them

(g) Explain why these *Aedes* mosquitoes are considered two species.

[3]

(h) Outline how the dengue virus infects immune cells in humans and spreads in the body.

[2]

In October 2016, the Health Sciences Authority (HSA) approved the use of Dengvaxia, the world's first dengue vaccine, in Singapore. The vaccine was recommended for people aged 12 to 45.

(i) Explain why the vaccine was not effective for people aged 46 and above.

[2]

Wolbachia is a diverse group of naturally occurring intracellular bacteria that infect and live in invertebrate cells and manipulate the biology of their hosts. Although they are found in more than 60 per cent of insect species, they are not found in the main vector of dengue *Aedes aegypti* in Singapore.

In order to suppress the mosquito population, researchers in Singapore conducted a field study that released fertile *Wolbachia*-carrying male mosquitoes into the urban built environment. This could greatly reduce the numbers of mosquitoes as the resulting zygote do not hatch.

Fig. 3.7 shows one possible effect of *Wolbachia* infection on the first zygotic mitosis.

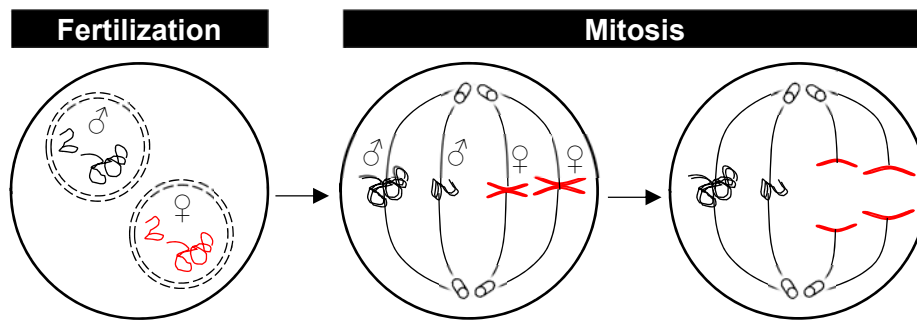


Fig. 3.7

- (j) (i) Explain why *Wolbachia* infection will be successful in reducing the number of mosquitoes over time.

[2]

Fig. 3.9 shows the food web in which mosquitoes are found in.

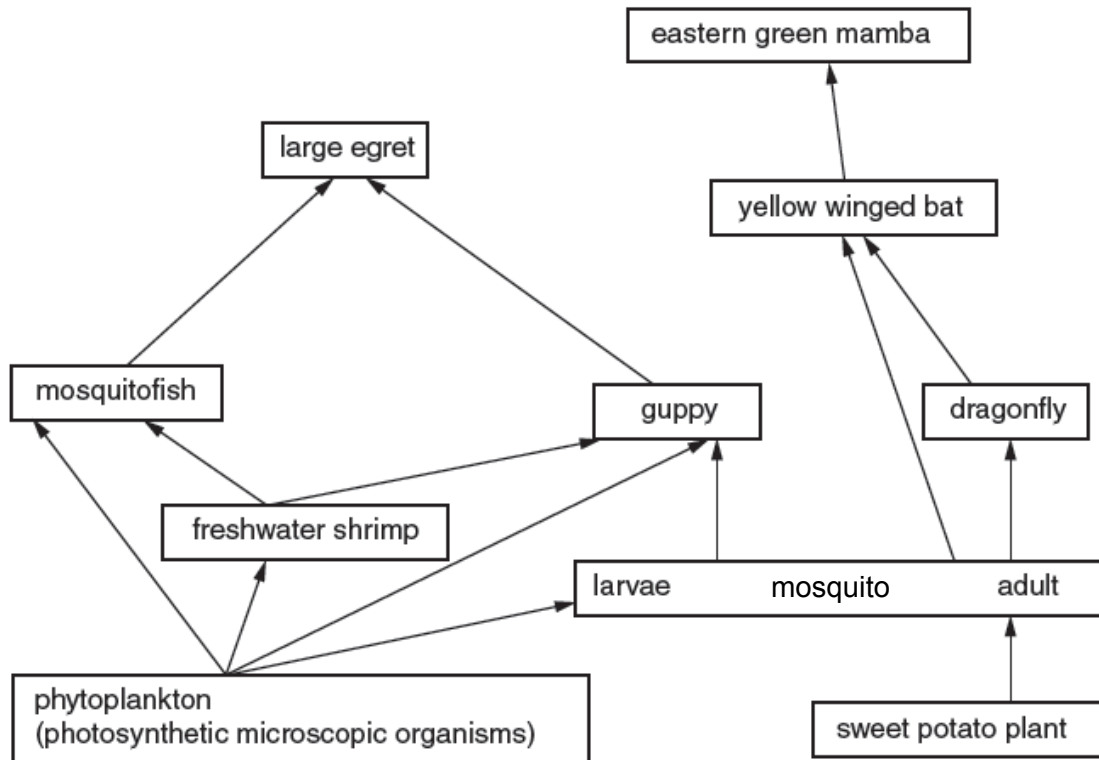


Fig. 3.9

- (ii) Explain which organism(s) found in this ecological niche will be greatly impacted by the release of *Wolbachia*-infected mosquitoes.

[2]

[Total: 28]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must set out in sections **(a)** and **(b)**, as indicated in the question.

EITHER

- 4** (a) Describe the mechanisms that give rise to the vast diversity of antibodies prior to antigen stimulation and describe how the antibody is formed from mature mRNA. [14]
- (b) Using a named disease, discuss how vaccination is an effective measure to control the disease. [11]

[Total: 25]

OR

- 5** (a) Describe the main stages of cell signaling and describe the roles of cAMP in eukaryotes and prokaryotes. [14]
- (b) Compare the signaling pathways between G protein coupled receptor and receptor tyrosine kinase in relation to blood glucose regulation. [11]

[Total: 25]

